

WILLOWMORE, South Africa

WMO: 688320

Lat: 33.3003S

Lon: 23.4839E

Elev: 2756

StdP: 13.29

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	34.0	17.0	14.4	61.7	20.6	17.1	59.4	21.2	56.4	19.1	57.6	1.0	240	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.5	95.1	63.7	91.5	63.2	88.3	62.3	68.0	83.1	66.8	81.5	65.5	79.8	8.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	99.6	72.0	62.6	94.1	71.2	61.0	88.7	70.7	34.0	82.8	32.9	82.1	31.9	79.7	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.6	16.3	14.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	26.4	70.2	1.7	1.8	25.2	71.5	24.2	72.5	23.2	73.6	22.0	74.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	72.9	71.8	69.1	61.9	56.4	49.7	49.8	52.7	56.9	62.1	64.8	69.7
	DBStd	10.13	5.85	5.74	5.75	7.08	5.88	5.38	6.04	6.30	6.23	7.44	6.52	5.81
	HDD50	217	0	0	0	4	14	68	77	41	12	1	0	0
	HDD65	2272	7	10	26	141	273	459	470	384	251	147	84	20
	CDD50	4390	709	610	591	360	211	59	72	125	220	377	444	612
	CDD65	971	252	200	152	47	5	1	0	3	9	57	78	167
	CDH74	9436	2436	1726	1224	469	81	10	12	76	166	789	798	1649
	CDH80	3840	1124	728	466	144	13	0	0	19	30	323	282	711
Wind	WSAvg	6.4	7.3	7.0	6.4	5.4	4.9	5.6	5.7	6.0	6.7	7.2	7.6	7.4
Precipitation	PrecAvg	10.4	1.1	1.3	1.5	1.2	0.7	0.5	0.5	0.6	0.4	0.7	1.1	1.0
	PrecMax	20.7	4.4	5.3	4.8	4.1	3.0	1.3	2.8	2.1	2.6	2.9	6.0	3.9
	PrecMin	3.9	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	3.8	1.1	1.4	1.1	0.9	0.8	0.4	0.6	0.6	0.6	0.7	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.6	98.2	94.0	89.2	81.7	75.0	75.6	82.1	84.4	94.9	93.0	98.0
		MCWB	64.6	66.0	63.5	60.0	57.7	53.1	53.0	54.3	56.6	61.7	60.6	62.9
	2%	DB	95.0	92.8	89.7	84.5	76.0	70.0	70.8	74.8	79.3	88.3	87.4	91.9
		MCWB	64.8	65.0	62.8	59.8	55.7	51.2	50.9	51.9	54.9	60.4	59.5	62.6
	5%	DB	90.4	88.3	85.4	79.3	71.7	65.7	66.8	69.6	74.4	82.6	82.9	87.6
		MCWB	63.9	64.3	61.9	59.0	54.5	49.8	49.8	50.3	53.3	58.3	59.0	61.8
	10%	DB	86.2	84.3	81.2	74.5	67.6	61.4	62.5	65.1	69.7	77.1	78.5	83.0
		MCWB	63.0	63.3	61.3	57.3	53.6	48.4	48.1	49.3	52.0	57.0	58.1	61.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	70.7	68.4	63.9	61.5	55.4	54.5	58.1	59.4	64.5	64.8	68.3
		MCDB	84.4	86.7	83.9	76.4	71.1	67.6	69.9	68.8	73.0	84.6	80.7	83.4
	2%	WB	67.4	68.4	66.1	62.4	58.2	52.9	52.8	55.5	57.0	62.6	62.6	66.1
		MCDB	83.2	83.2	80.5	77.6	69.1	64.8	66.1	66.4	73.0	81.5	77.7	82.5
	5%	WB	66.4	67.1	64.7	60.9	56.6	51.2	51.3	53.3	55.2	60.6	60.8	64.3
		MCDB	82.1	80.6	77.7	73.9	67.7	62.4	62.5	62.9	69.5	75.5	77.8	80.3
	10%	WB	65.1	65.7	63.2	59.2	55.0	49.6	49.5	51.2	53.4	58.6	59.5	62.9
		MCDB	80.6	77.7	76.0	71.0	65.6	59.5	59.4	62.1	66.5	73.3	74.8	77.6
Mean Daily Temperature Range	5% DB	MDBR	24.5	23.1	21.9	21.5	20.9	20.6	21.6	21.9	22.2	23.1	23.3	24.7
		MCDBR	32.0	29.9	29.6	28.9	27.3	27.7	28.4	29.8	31.3	33.9	31.3	32.6
		MCWBR	8.6	7.9	8.0	8.9	11.2	12.8	12.6	12.3	10.9	10.5	8.3	9.1
	5% WB	MCDWR	26.8	24.8	24.2	23.9	23.2	22.3	23.1	22.5	27.2	28.9	27.5	28.0
		MCWBR	8.5	7.5	7.0	8.3	10.5	11.4	11.7	9.9	10.8	9.9	8.3	9.2
Clear-Sky Solar Irradiance	taub		0.333	0.338	0.325	0.300	0.284	0.273	0.275	0.302	0.321	0.337	0.323	0.329
	taud		2.532	2.517	2.542	2.602	2.624	2.632	2.614	2.512	2.443	2.433	2.521	2.531
	Ebn at Noon		319	311	304	295	284	280	285	289	300	307	320	321
	Edh at Noon		35	34	31	26	23	21	23	28	34	37	35	35
All-Sky Solar Radiation	RadAvg		2423	2174	1790	1363	1041	899	984	1252	1671	2015	2340	2512
	RadStd		93	81	110	61	46	43	60	68	90	110	124	156

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page